

PRANSHU SINGLA

+91-7710272011 | pranshusingla662@gmail.com | GitHub | LeetCode | LinkedIn

SUMMARY

Innovative Software Engineer specializing in Full Stack Development and AI Integration. Proven expertise in building scalable web applications using Next.js and MongoDB, enhanced by intelligent features via LLMs (Gemini API) and RAG architectures. Strong foundation in Data Structures and Algorithms (C++) with a track record of optimizing application performance and increasing user engagement by up to 50%. Passionate about bridging the gap between complex AI models and intuitive user interfaces.

EDUCATION

National Institute of Technology, Delhi
B.Tech, Computer Science & Engineering • **GPA: 8.00**

Aug 2024 – May 2028 (Expected)
New Delhi, India

TECHNICAL SKILLS

- **Languages:** Python, C++, C, JavaScript, TypeScript, SQL
- **AI / ML & GenAI:** Generative AI, LLMs, LangChain, LangGraph, RAG Pipelines, Computer Vision, Deep Learning, Machine Learning, Scikit-Learn, PyTorch, YOLOv8, OpenCV, Albumentations, Gemini API, Anthropic Claude API
- **Web & Backend:** Next.js, React.js, Node.js, FastAPI, REST API, MongoDB, PostgreSQL, Prisma, Clerk, Stream, Redis, NextAuth, Tailwind CSS
- **Tools & Platforms:** AWS, Vercel, Render, Docker, Git, GitHub
- **Databases & Infra:** MongoDB Atlas, PostgreSQL, Redis, Mongoose ODM, DBMS

PROJECTS

ProTrack — Smart Classroom Project Management System

May 2026

- Next.js 14, MongoDB Atlas, NextAuth.js v5, Gemini API, Tailwind CSS, shadcn/ui, Zod, Cloudinary, Vercel, TypeScript, React 18, Node.js, Google OAuth 2.0
- ▶ Architected a **full-stack, cloud-hosted academic platform** (Next.js 14 App Router + MongoDB Atlas) managing the complete project lifecycle — class creation → group formation → rubric-based evaluation → grade publication — for NIT Delhi's CSE department.
 - ▶ Integrated **Gemini API** to build an **AI Architect** (auto-generates project structures, rubrics, milestone plans from natural-language prompts) and a **Professor AI Chatbot**, making it one of the first LLM-augmented academic management platforms.
 - ▶ Implemented a **5-layer security architecture**: Google OAuth 2.0, HTTP-only JWTs, **RBAC middleware**, Zod runtime validation, and full OWASP Top-10 compliance; achieved sub-300ms P95 response times under 100 concurrent users in Apache JMeter load tests.
 - ▶ Scored **SUS 81.4 / 100 ('Excellent')** in UAT with 12 NIT Delhi participants; zero high/critical vulnerabilities found in OWASP ZAP penetration testing.

JobFinder — AI-Agent Recruitment Platform

Jan 2026

- Next.js, FastAPI, LangGraph, LangChain, Gemini 1.5, MongoDB, Scikit-Learn, Tailwind CSS, Framer Motion, JWT, NextAuth, Google OAuth 2.0, Python
- ▶ Built a micro-services **AI-agent** recruitment platform with a **LangGraph** autonomous agent capable of executing live MongoDB queries, generating tailored cover letters, and submitting job applications via tool-calling — end-to-end without human intervention.
 - ▶ Designed a two-pass candidate ranking engine combining MongoDB text-search heuristics with LLM semantic scoring (Gemini 1.5 Flash), delivering AI-summarized talent matches with significantly reduced API latency vs. single-pass LLM approach.
 - ▶ Engineered glassmorphism UI with RBAC, Google OAuth, and real-time analytics dashboard; Scikit-Learn ML model integrated for automated resume-to-job relevance scoring.

RadhaInf — Fog-Robust Vehicle Detection System -- Improved

Dec 2025

- PyTorch, YOLOv8, OpenCV, Albumentations, Flask, Python, Image Dehazing, Computer Vision
- ▶ Developed a fog-robust vehicle detection model using **YOLOv8 with custom fog-aware augmentation pipeline** (Albumentations) and image dehazing preprocessing; achieved **mAP@0.5 = 0.75** and mean **recall 73%** on an unseen foggy test set.
 - ▶ Outperformed internal clear-image baseline by **+34% mAP@0.5** and **+18% average recall** in foggy conditions; built unified inference pipeline supporting images, video files, webcam, and Android IP camera streams with real-time dehazing + detection.

ACHIEVEMENTS & CERTIFICATIONS

- ▶ Qualified for Pre-Final Round of Industrial Ideathon by DSIIDC 2025 — Top 100 teams out of 1,700+ teams across Delhi NCR.
- ▶ Selected as Delegate from NIT Delhi to attend National Workshop at IIT Kanpur.
- ▶ Improved breast-ultrasound segmentation accuracy on the BUSI dataset from 86% → 96% (+10%) through hyperparameter tuning, reducing model error by 5%.
- ▶ Completed AWS Cloud Practitioner Essentials (Forage).
- ▶ Appointed as Hackerrank Campus Ambassador from NIT Delhi.
- ▶ Solved 300+ problems on Leetcode and have a rating of 1030 on codeforces and 1684 on Leetcode.